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Digital Optimization of Marine Transport Through an Integrated Barge Cycle Management System (BCMS)

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Abstract

Marine construction projects demand the continuous movement of large volumes of aggregates from quarries to offshore construction sites, where efficiency in material transportation is critical to meeting tight schedules and controlling costs. Despite the operational complexity of this supply chain, reporting and tracking processes often remain manual, leading to fragmented data, limited visibility, and slow decision-making. These inefficiencies frequently manifest as cycle delays, underutilized resources, and escalating costs, highlighting a gap in sector-specific digital tools designed for marine logistics. This paper presents the Barge Cycle Management System (BCMS) developed to address this gap through a fully integrated, location-based digital platform that manages the entire barge cycle— from production and inspection to loading, sailing, offloading, and maintenance. BCMS standardizes reporting through modular templates and validates inputs using a multi-layer verification engine, ensuring accurate, traceable data and resolving inconsistencies before they affect operations. A live deployment demonstrated improved cycle transparency, real-time bottleneck detection, and actionable insights for optimizing tug-barge pairings and fleet allocation. Beyond immediate operational benefits, BCMS provides a scalable foundation for advanced analytics, long-term trend monitoring, and future integration with predictive modeling and marine traffic systems. This work demonstrates how targeted, problem-specific digital transformation can enhance efficiency and transparency in marine material transport, supporting the transition toward more data-driven construction practices.

Introduction

Efficient transport of rock aggregates from quarries to offshore construction sites forms the logistical backbone of marine infrastructure projects. Despite the operational scale and complexity of this material flow, the marine sector—particularly in the domain of marine transport and cycle reporting—has been slow to adopt comprehensive digital solutions. Traditional reporting methods rely heavily on manual data collection, fragmented submissions from multiple stakeholders, and post-event reconciliation, which results in poor visibility, delays in detecting critical issues, and limited decision-making capacity. Moreover, this lack of integration across teams and phases often results in data discontinuity, inefficient fleet utilization, and reactive management approaches that impact project timelines and costs. Additionally, the absence of structured and real-time insights hinders long-term analysis of performance trends and supply chain optimization, limiting opportunities for continuous improvement.

To address these challenges, the **Barge Cycle Management System (BCMS)** was developed as a modular, end-to-end logistics management platform that digitizes the full barge transport cycle. Designed specifically for marine logistics, BCMS decentral izes data entry to operational stakeholders while enforcing sequence validation, time continuity, and material compliance checks across the barge lifecycle—from production and loading to sailing, offloading, and maintenance. By transforming raw operational data into a structured, validated, and analytics-ready format, the system delivers both immediate operational visibility and long-term strategic value for marine logistics management.

State of the Art Review

Studies have extensively examined delays and cost overruns in marine construction projects. The World Bank notes that delays in transportation infrastructure significantly escalate construction costs [1], highlighting the importance of sustainable solutions that can help minimize project risks. Majority of the transportation involves construction materials. When it comes to material management, this also underscores the importance of accurate marine movement tracking and reliable data collection for timely decision-making related to cost and time. These decisions are highly dependent on a number of factors including marine vessels and other equipment, material being transported, and logistical management. Project management techniques play an important role in having a complete organized project which covers managing the materials, machinery, budget, and methods used [2]. This highlights a growing need for improved systems that can streamline these management processes by integrating monitoring, analytics, and reporting to enhance overall project efficiency and reduce the risks of cost overruns and schedule delays.

Tam and Shen identify weather, equipment, and logistical uncertainties as major risks in marine construction and propose structured frameworks for risk assessment and mitigation [3]. Their work highlights the value of proactive risk strategies in minimizing time and cost overruns. This again highlights the need for clear visibility on the project's status, especially concerning the marine movements, to enable efficient decision-making. Similarly, Shrivas and Singla analyze delays in Indian marine construction projects, emphasizing weather, logistical, and managerial inefficiencies as key contributing factors [4]. While weather impacts are unavoidable during operations, managerial and logistical delays can be significantly mitigated through sophisticated monitoring and decision-support systems like BCMS. Moreover, such systems can incorporate weather trend analyses, enabling better long-term scheduling and contingency planning for marine operations.

More recently, digitalization trends in marine logistics have shown promising results. An ADIPEC 2023 study demonstrated realtime digital solutions that improved tracking, data integration, and operational efficiency [5]. Broader research on AI-driven digital transformation highlights its potential to create a more transparent and productive marine transportation sector [6].

Despite recent advancements, many marine civil processes remain largely manual, causing significant delays in data collection and decision-making, key contributors to cost and schedule overruns. This stems from the sector's reliance on generic digital tools that fail to capture the unique complexities of marine logistics, such as barge cycle optimization, marine traffic coordination, predictive maintenance, and material tracking. Although existing literature provides valuable insights into the causes of delays and highlights the potential of digitalization and AI-driven tools, most studies present broad frameworks rather than addressing the specific operational cycles that drive marine logistics performance. The Barge Cycle Monitoring System (BCMS) addresses this gap by focusing on the marine material transportation cycle—a critical yet underrepresented aspect of project efficiency—while being modular enough for future integration into larger digital ecosystems.

Methodology

The transportation of rock aggregates from quarry facilities to offshore project sites is a coordinated process involving multiple operational components working in parallel. The cycle begins with the generation of material requests from the project site, based on construction schedules and daily work plans. These requests are communicated to the quarry, which begins preparing the required material quantities accordingly. Material production, testing, and quality inspections occur as part of a continuous process at the quarry. Once aggregates are produced, they are tested and, upon approval, added to the available stockpile. This production-approval workflow runs independently and may not always be synchronized with barge dispatch operations. Approved stock can accumulate in anticipation of future transport, allowing flexibility in scheduling barge movements.

The barge transport cycle itself is initiated when a tug-barge unit is assigned for material delivery. The process starts with loading the approved aggregates onto the barge at the quarry jetty. After loading, the barge is connected to a tug and departs for the offshore site. Upon arrival, the material is offloaded, and the empty barge returns to the quarry to await the next dispatch. Between barge trips, scheduled maintenance activities may be carried out, or unplanned mechanical issues may be addressed as needed. These events are typically recorded manually and can impact overall productivity and cycle timing. The BCMS has digitalized this entire process to achieve an optimized process throughout the project duration.

System Design

The Barge Cycle Management System (BCMS) was developed to digitalize and optimize the material transport process between quarries and offshore project sites. The system's architecture is designed around modular, location-based reporting that mirrors the real-world operational flow while introducing structure, validation, and analytical capabilities. BCMS integrates data from production, loading, sailing, offloading, and maintenance operations into a unified platform, enabling real-time visibility and data-driven decision-making across the marine logistics chain. Fig 1 shows the modular structure of the BCMS, highlighting key features.

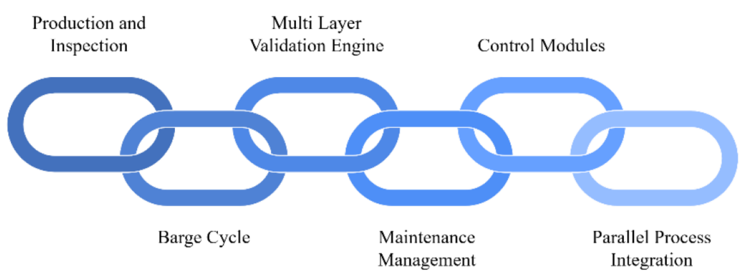


Figure 1—System Modules

Production and Inspection. This production and inspection stage is independent of barge movement. Each day, a dedicated team at each quarry reports the production of materials as requested by the project. The produced material is then inspected by the quality assurance teams using various testing methods. The tested and approved quantities are reported along with their corresponding batch and reference numbers. Once approved, the quantities are added to the stockpile and become ready for transportation. This process ensures clear visibility of material availability between the quarry and project sites, enabling effective barge cycle planning and better management of contractual terms with material suppliers.

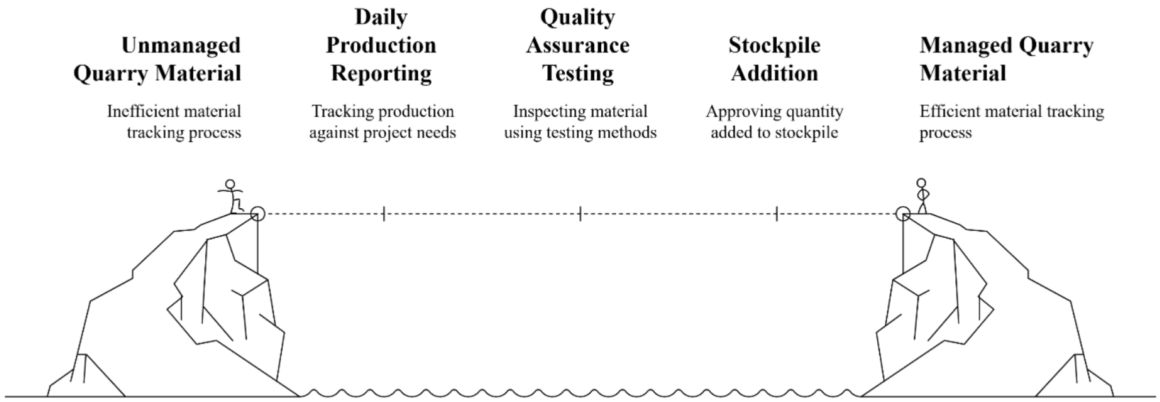


Figure 2—Managed Quarry

Barge Cycle. The second module consists of the barge cycle shown in Fig. 3. The cycle is split into the following phases.

- Phase 1: Loading
- Phase 2: Sailing loaded to site
- Phase 3: Offloading at site
- Phase 4: Sailing empty
- Phase 5: Maintenance (Planned)

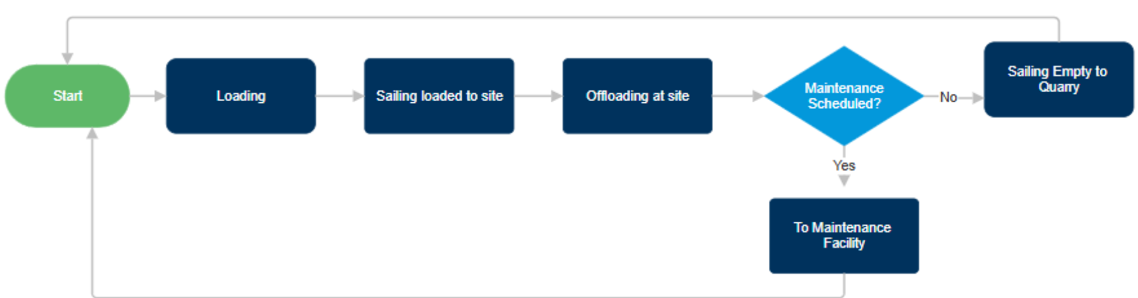


Figure 3—Barge Cycle Process flow

The barge cycle starts from loading at the quarry, sailing loaded to the site, offloading material at the site, and depending on maintenance plans, either sailing to the maintenance facility or back to the loading point. Since the entire process involves multiple teams, such as project sites, supplier quarries, and marine logistics, the system ensures that the reporting follows a sequential approach while providing each team the ability to add their part of the report within the cycle. This gives visibility into the current stage of the cycle, along with dedicated accountability between teams to ensure the entire cycle is closed and traceable.

Sequential Reporting Enforcement

To ensure data continuity and traceability, the Barge Cycle Management System (BCMS) enforces a strict sequential logic for reporting each barge cycle. A new operational phase (e.g., *Sailing Loaded*) cannot be submitted unless the preceding phase (e.g., *Loading*) has been recorded and validated. This mechanism prevents data gaps, overlaps, or out-of-sequence entries, thereby ensuring a complete and auditable barge journey. Every cycle is uniquely identified by a **cycle number**, which acts as a reference key across different reporting stages. This identifier allows decentralized teams to contribute reports while maintaining system-wide consistency.

Phase-wise Validation Logic

- **Phase 1 (Loading):** When a report for phase 1 is submitted, the system checks if the barge's previous cycle has ended in phase 4 (Sailing Empty) or phase 5 (Maintenance). If this condition is met, the system closes the previous cycle and initiates a new one. It then verifies data continuity and accepts the phase 1 report. If no prior cycle exists, the system verifies whether phase 1 data already exists for the current cycle. If it does, continuity is checked before updating; otherwise, a new cycle is started.
- **Phase 2 (Sailing Loaded):** The system accepts a phase 2 report only if the previous report belongs to phase 1 or phase 2. If the previous report is from phase 1, it closes phase 1, performs continuity validation, and records phase 2. If the previous report is already from phase 2, only continuity validation is performed before data entry.
- **Phase 3 (Offloading):** A phase 3 report is accepted only if the previous report belongs to phase 2 or phase 3. If the previous report is from phase 2, the system closes that phase, validates continuity, and logs phase 3. If the previous report is from phase 3, it performs continuity checks before accepting the new data.
- **Phase 4 (Sailing Empty):** Phase 4 can only follow phase 3 or another phase 4 entry. If the previous entry is phase 3, the system closes it, checks continuity, and records phase 4. If the previous entry is from phase 4, the system proceeds with continuity validation before uploading the new data.
- **Phase 5 (Maintenance):** The system accepts a phase 5 report only if the prior phase is 4 or 5. If the preceding entry is from phase 4, the system closes both phase 4 and the entire cycle before recording the maintenance phase, subject to continuity checks. If the previous entry is already phase 5, only continuity validation is performed.

Each responsible team has controlled access to their respective sections of the reporting workflow, meaning they can submit and edit only the phases assigned to them. However, they have read-only visibility of reports submitted for other phases within the same cycle. This transparency fosters open communication and cross-team collaboration, allowing discrepancies to be quickly identified and resolved while maintaining data integrity and accountability.

Standardized decentralized phase activities

To standardize the reporting of each phase, a structured reporting module with predefined activities was developed following extensive discussions with multiple stakeholders and teams, shown in Table 1. This standardization aimed to reduce reporting ambiguities, enhance data quality, and ensure traceability of vessel performance across all operational phases.

Table 1—Standardized Activities

Operational			Non-Operational		
Main Activity	**Sub Activity**	**Sub-Sub Activity**	**Main Activity**	**Sub Activity**	**Sub-Sub Activity**
Operational (Phase 1)	Loading		Non-Operational (All Phases)	Off hired	
Operational (Phase 2)	Sailing Loaded			Dead Time	Weather
Operational (Phase 2 & 4)	Berthing				Client Delay
	Unberthing				Awaiting Berth
Operational (Phase 3)	Offloading	Side Dumping		Technical Delays	Operational Delay
		Dry Dumping/Tipping			Breakdown
		Marine Installation			Tug Maintenance
		Offloading At Jetty		Other Delays	Barge Maintenance
Operational (Phase 4)	Sailing Empty				Port Delay
	Sailing to Maintenance				No Tug
Operational (Phase 5)	Barge Maintenance	Deck Concreting			No Pilot
		Side Walls Fabrication			No EME/Operator
		Inspection			
		General Maintenance			
	Tug Maintenance	Major Engine Overhaul			
		Inspection			
		General Maintenance			
	Standby	Survey Activities			
		Standby			

Multi-Layer Validation Engine. At the core of the Barge Cycle Management System (BCMS) lies a robust multi-layered validation engine, designed to ensure that all reported data adheres to strict temporal, logical, and operational rules before it is accepted. This engine plays a pivotal role in maintaining data integrity, enabling reliable analytics, and supporting informed decision-making across marine logistics operations.

General Validation Rules

The General validation engine performs a series of foundational checks applicable to all phases of the reporting workflow:

- **Timestamp validation:** Ensures that reported events follow a correct chronological sequence (e.g., loading must occur before sailing, which must occur before offloading). This maintains the temporal accuracy of each barge trip.
- **Activity validation:** Ensures that each phase is reporting with its phase-specific activities.
- **Continuity check:** Detects missing, duplicated, or out-of-sequence entries within a phase and between phases.

Phase-Specific Validation Rules

In addition to general validations, the engine enforces context-aware checks tailored to specific operational phases. These rules ensure that data reported during each phase is not only accurate but also meaningful within its operational context.

Loading and Offloading Phases

- Material type and quantity are mandatory fields if the barge is operational.
- Berth information must be specified to track the loading/offloading location and improve berth scheduling efficiency.
- For projects segmented into multiple offloading areas, the report must include a breakdown of the quantity offloaded per location, ensuring traceability and proper distribution across work zones.

Sailing Phase

- A tug must always be linked to the barge to validate vessel movement.
- Sailing speed must be reported to assess transit efficiency and tug performance.

In summary, the multi-layer validation engine not only enforces baseline data quality but also promotes phase-aware precision. Fig 4. shows the system flagging a sailing entry without a tug linked with the barge.

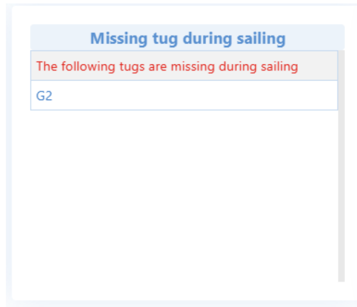


Figure 4—Error flagged while uploading the sailing report.

Maintenance Management. While scheduled maintenance and downtimes are logged outside of a cycle. Any unplanned maintenance and breakdowns are logged within the cycle, under the sub-activity 'Technical delays', resulting in a higher cycle duration. This approach clearly shows the outliers in the data and the delays caused due to unexpected reasons.

Control Module. The Control Module of the BCMS offers comprehensive tools for setting up projects and suppliers, aiming to digitally replicate real-world project and quarry environments. This ensures that reporting remains as accurate and reflective of on-ground operations as possible. Additionally, the module includes functionality to upload and manage Key Performance Indicators (KPIs), enabling real-time monitoring of both equipment efficiency and overall project performance.

The KPI control module provides the option to add loading and offloading rates for each material type, along with offloading method, average sailing time and distance between each quarry and project site, overall project quantities for each material, estimated berthing and unberthing times as well as a quick cycle time estimate for different materials given the barge capacity as shown in Fig. 5 (a). The General Setup module provides the option to add project and supplier jetty, berths, loading/offloading equipment, project areas to manage offloading splits, and maintenance facilities to track the barge location during maintenance, shown in Fig. 5 (b).

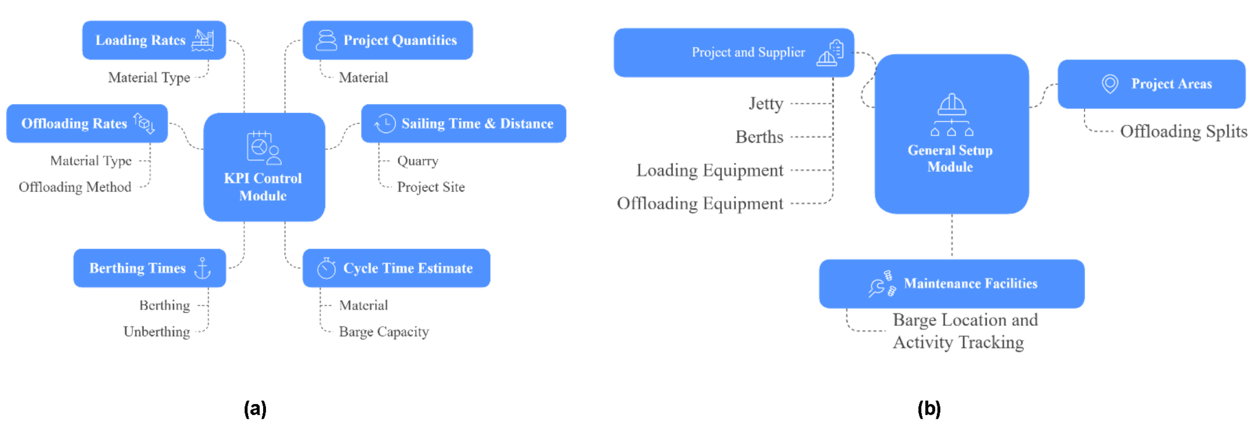


Figure 5—Control Modules (a) KPI control Module; (b) General Setup Module

Parallel Process Integration. The BCMS distinguishes between material production and barge cycles as two parallel yet interconnected processes. While these operations run independently, the system ensures full real-time visibility across both. Approved material stockpiles are continuously tracked. Once a barge is loaded, the loaded quantity is automatically subtracted from the corresponding stockpile, maintaining accurate and up-to-date material availability data. This approach ensures operational flexibility, prevents dispatch delays due to stock shortages, and maintains seamless coordination between production and logistics. The system logic links the two workflows at the loading checkpoint, ensuring automated updates to the stockpile.

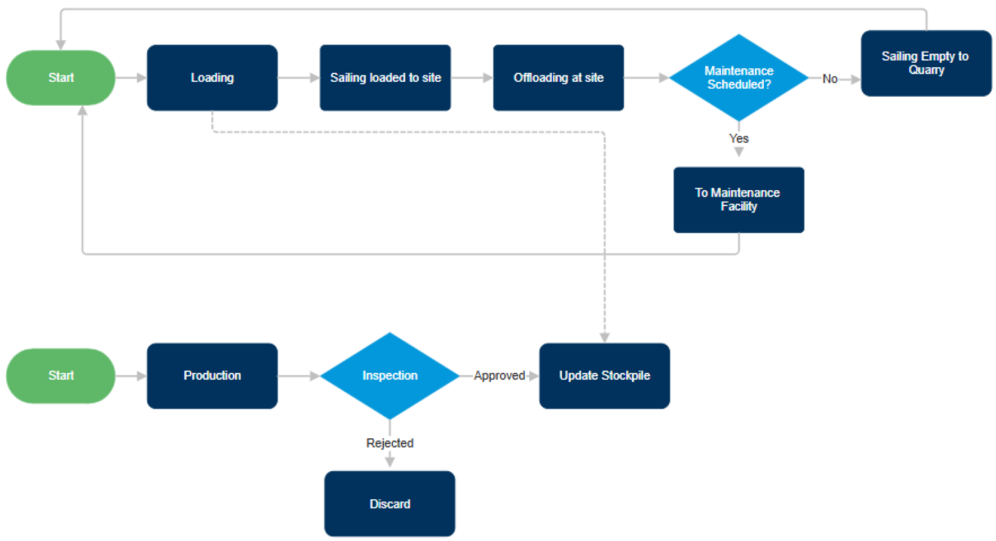


Figure 6—Integration of the Barge Cycle process and material production and inspection process.

System Outputs and Optimization Approach. Validated data is visualized through dashboards and fed into pipelines for further analytics such as daily tracking, delay analysis, berth and site utilization, material throughput and compliance, and finally a complete overview of the cycle time deviations and trends. This supports both real-time operational adjustments and long-term strategic planning. The optimization logic relies on minimizing cycle idle times and maximizing material throughput. This is achieved by continuously comparing planned vs. actual loading, sailing, and offloading durations, and flagging deviations for corrective action. The framework also prioritizes resource reallocation based on vessel's cost vs performance metrics ensuring more efficient deployment of available assets.

Results and Analysis

To monitor the results, the BCMS was initially implemented in one of our projects before rolling it out to all project sites. We selected a project with a total of 25 barges mobilized over different periods of time throughout the project. The project required 10 different types of rocks to be delivered. The results were monitored over a period of 4 months. The qualitative results were based on the feedback from the site teams, while the quantitative analysis followed certain evaluation metrics.

Evaluation Metrics

To evaluate the results, a certain set of estimates and KPIs was used, where estimated cycle time remains one of the most important evaluation parameters.

$$Est.Cycle\ Time\ [hours] = \frac{Total\ Barge\ Capacity}{Loading\ Rate} + \frac{Total\ Barge\ Capacity}{Offloading\ Rate} + Total\ Sailing\ Time + Total\ Berthing\ Time$$

$$Total\ Sailing\ Time\ [hours] = Estimated\ Saling\ Loaded\ Time + Estimated\ Sailing\ Empty\ Time$$

$$Total\ Berthing\ Time\ [hours] = (2*\ Estimated\ Berthing\ Time\ ) + (2*\ Estimated\ Unberthing\ Time$$

$$Loading\ Rate\ [TPH] = \frac{Material\ Quantity\ Loaded\ (Tonn)}{Loading\ Hours}$$

$$Offloading\ Rate\ [TPH] = \frac{Material\ Quantity\ offloaded\ (Tonn)}{Loading\ Hours}$$

Cycle KPI %: The percentage of trips that were completed within the estimated cycle time.

$$Cycle\ KPI\ \% = \frac{No.of\ Cycles\ on\ time}{Total\ No.of\ Cycles} * 100$$

$$No.\ of\ \ Cycles\ on\ time = Cycles\ with\ Total\ time < Est.Cycle\ Time$$

Dead Time %: The total duration barges spent without working for several reasons.

$$Dead\ Time\ \% = \frac{Dead\ Time}{Total\ time} * 100$$

Awaiting Berth %: The percentage of berth awaiting time out of the total dead time.

$$Awaiting\ Berth\ \% = \frac{No\ Berth\ Time}{Dead\ Time} * 100$$

$$Awaiting\ Berth\ at\ quarry\ \% = \frac{No\ Berth\ Time\ (phase=1)}{Dead\ Time} * 100$$

$$Awaiting\ Berth\ at\ site\ \% = \frac{No\ Berth\ Time\ (phase=3)}{Dead\ Time} * 100$$

Table 2. shows the loading and offloading estimates in TPH.

Table 2—Loading and offloading rates

Loading/Offloading	Quarry Run [TPH]	Under Layer [TPH]	Armour [TPH]
Loading	300	300	500
Offloading – Side Dumping	500	300	
Offloading – Marine Installation	400	700	600
Offloading At Jetty	500	500	600

Table 3. shows the average sailing and berthing/unberthing times in hours, based on the distances between the project site and quarry.

Table 3—Average sailing and berthing/unberthing time

Activity	Duration [hours]
Sailing Loaded	88
Sailing Empty	88
Berthing	20
Unberthing	20

Daily Tracking and Visibility

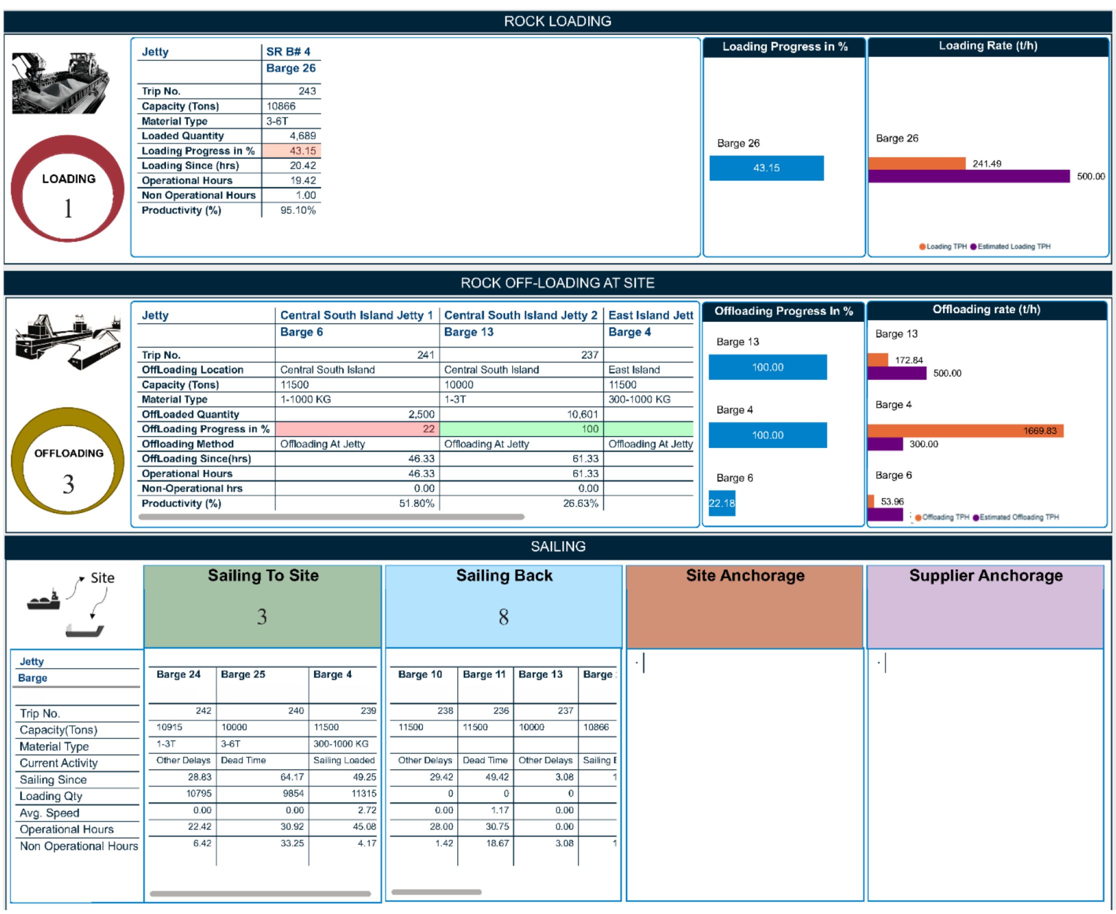
One of the most immediate and impactful outcomes of implementing the Barge Cycle Management System (BCMS) was the ability to achieve real-time operational visibility. Through the system's live dashboard, stakeholders are now able to monitor the current status of every barge involved in active operations. This includes the number of barges currently loading at quarries, sailing to site, offloading at project locations, or returning empty.

This daily tracking feature enables:

- Instant situational awareness for operational coordinators
- Faster decision-making in response to delays or bottlenecks
- Improved planning of loading/offloading slots and tug allocations
- Minimized idle time by proactively managing upcoming phases
- Live information of the available stockpile at the quarry to schedule cycles

At the loading point, the dashboard compares the actual loading rate against the estimated rate, automatically flagging any deviations where actual performance falls short. It also tracks loading progress relative to total barge capacity and displays the number of barges waiting at the quarry, helping teams optimize turnaround time. At the offloading point, the system highlights barges with reduced offloading rates, enabling immediate intervention to address underlying issues. It also indicates berth occupancy and queue status, supporting better berth allocation and reduced congestion. Finally, it also shows offloading progress based on the total loaded material. During the sailing phase, the dashboard displays the sailing speed of each tug-barge combination along with the number currently en route to or from the site. This allows teams to plan based on the projected arrival of barges, optimizing site readiness and offloading preparations.

With full visibility across each operational phase, project teams are able to identify bottlenecks in real-time and respond swiftly, improving overall site responsiveness and coordination. Fig 7. shows the snapshot of the dashboard.



Jetty	Central South Island Jetty 1	Central South Island Jetty 2	East Island Jett
	Barge 6	Barge 13	Barge 4
Trip No.	241	237	
OffLoading Location	Central South Island	Central South Island	East Island
Capacity (Tons)	11500	10000	11500
Material Type	1-1000 KG	1-3T	300-1000 KG
OffLoaded Quantity	2,500	10,601	
OffLoading Progress in %	22	100	
Offloading Method	Offloading At Jetty	Offloading At Jetty	Offloading At Jetty
OffLoading Since(hrs)	46.33	61.33	
Operational Hours	46.33	61.33	
Non-Operational hrs	0.00	0.00	
Productivity (%)	51.80%	26.63%	

Jetty / Barge	Barge 24	Barge 25	Barge 4	Barge 10	Barge 11	Barge 13	Barge
Trip No.	242	240	239	238	236	237	
Capacity(Tons)	10915	10000	11500	11500	11500	10000	10866
Material Type	1-3T	3-6T	300-1000 KG				
Current Activity	Other Delays	Dead Time	Sailing Loaded	Other Delays	Dead Time	Other Delays	Sailing I
Sailing Since	28.83	64.17	49.25	29.42	49.42	3.08	1
Loading Qty	10795	9854	11315	0	0	0	
Avg. Speed	0.00	0.00	2.72	0.00	1.17	0.00	
Operational Hours	22.42	30.92	45.08	28.00	30.75	0.00	
Non Operational Hours	6.42	33.25	4.17	1.42	18.67	3.08	1

Figure 7—System Output: Dashboard showing live tracking for quick decision-making

Delay Analysis

As part of the delay analysis, the major delay categories were monitored. Table 4. Shows the major delays and their percentages. As dead times were significantly higher than the remaining delays, it was evident to look into this category to find out the problem.

Table 4—Major Delay Reasons

Delay	Percentage
Dead Time	35.18
Other Delays	2.09
Technical Delays	0.78

A detailed investigation into the data indicated that the primary cause of these delays was berth unavailability, as shown in Table 5. Supplier berths, though partially outside operational control, frequently lacked timely availability. Project berths, on the other hand, showed potential for optimization in allocation and scheduling. Weather-related delays were another significant factor, underscoring the need for more proactive planning. One potential mitigation strategy is to schedule planned maintenance activities during forecasted weather downtime; however, the impact of seasonal weather variations cannot be fully eliminated.

Table 5—Breakdown of the total dead time into for deeper analysis

Delay Sub-Activity	Percentage
Awaiting Berth	18.24
Weather	12.45

A comprehensive breakdown of the delays is presented in Table 6, with the third major category identified as *operational delays*. Detailed analysis of reporting remarks indicated that these delays primarily stemmed from activities such as shifting barges between berths, awaiting clearance to sail, and relocating offloading equipment. Although these delays were relatively minimal, they could be further reduced through improved coordination among site operational teams.

Table 6—All delay categories and their percentages

Delay Sub-Activity	Percentage
Awaiting Berth	18.24%
Weather	12.45%
Operational Delay	4.59%
General Maintenance	2.59%
No Tug	1.37%
Inspection	1.18%
Port Delay	0.70%
Barge Maintenance	0.49%
Standby	0.42%
Tug Maintenance	0.11%
Deck Concreting	0.10%
Major Engine Overhaul	0.10%
Breakdown	0.09%
Client Delay	0.04%

The findings suggest that inefficiencies in berth management contributed significantly to increased turnaround times. Streamlining project berth allocation is therefore identified as the most critical opportunity to reduce delays and associated costs.

To assess the impact on average cycle time, berth awaiting times were excluded from the calculation. The monitored cycle time shown in Fig 8. showed an average of 0.802 days (19.2 hours) in April, which decreased to 0.749 days (17.9 hours) in July, representing an approximate reduction of 6.6 %. The percentage is calculated as per the equation below.

$$Cycle\ Time\ \mathrm{Reduction}\ \% = \frac{Previous\ Cycle\ Time - Current\ Cycle\ Time}{Current\ Cycle\ Time} * 100$$

$$Cycle\ Time\ \mathrm{Reduction}\ \% = \frac{0.802 - 0.749}{0.802} * 100$$

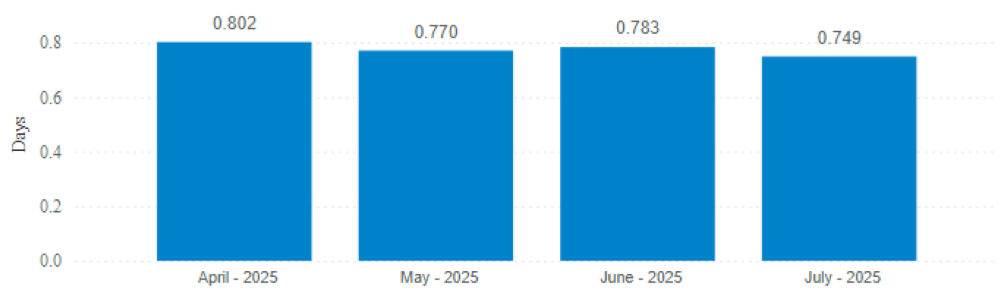


Figure 8—Average Cycle Time (in days) Excluding Berth Awaiting

Cost VS Performance Analysis

As part of the system's analytical capabilities, a cost versus performance analysis was conducted for each tug. In this analysis, cost was represented by the vessel's daily rental rate, while performance was measured using the average sailing speed recorded during active operations, specifically when the tug was either sailing to the site or returning to the quarry.

This comparative assessment provided valuable insights into the operational efficiency of high-cost versus low-cost assets, helping identify underperforming vessels and make informed decisions regarding asset utilization and hiring strategies.

Based on the plotted analysis shown in Fig 9. three tugs fell into the high-cost/low-performance quadrant, signaling inefficiencies that prompted immediate internal review and discussion for further actions, including replacement and contractual re-negotiations.

Figure 9—Cost vs performance analysis of tugs

Trend Analysis

A trend analysis was conducted over a period of four months to evaluate improvements in total cycle times and associated delays. The objective was to monitor the percentage of cycles completed within the targeted time frame every month, using Cycle KPI percentage plotted over time. Initial findings shown in Fig. 10 showed a notable improvement, with cycle achievement rising from approximately 51.06% in April to 82.75% in June. However, as more barges were deployed, increased berth congestion—particularly at loading and offloading points—emerged as the major factor impacting performance, leading to a decline in overall cycle efficiency to 78.92% in July.

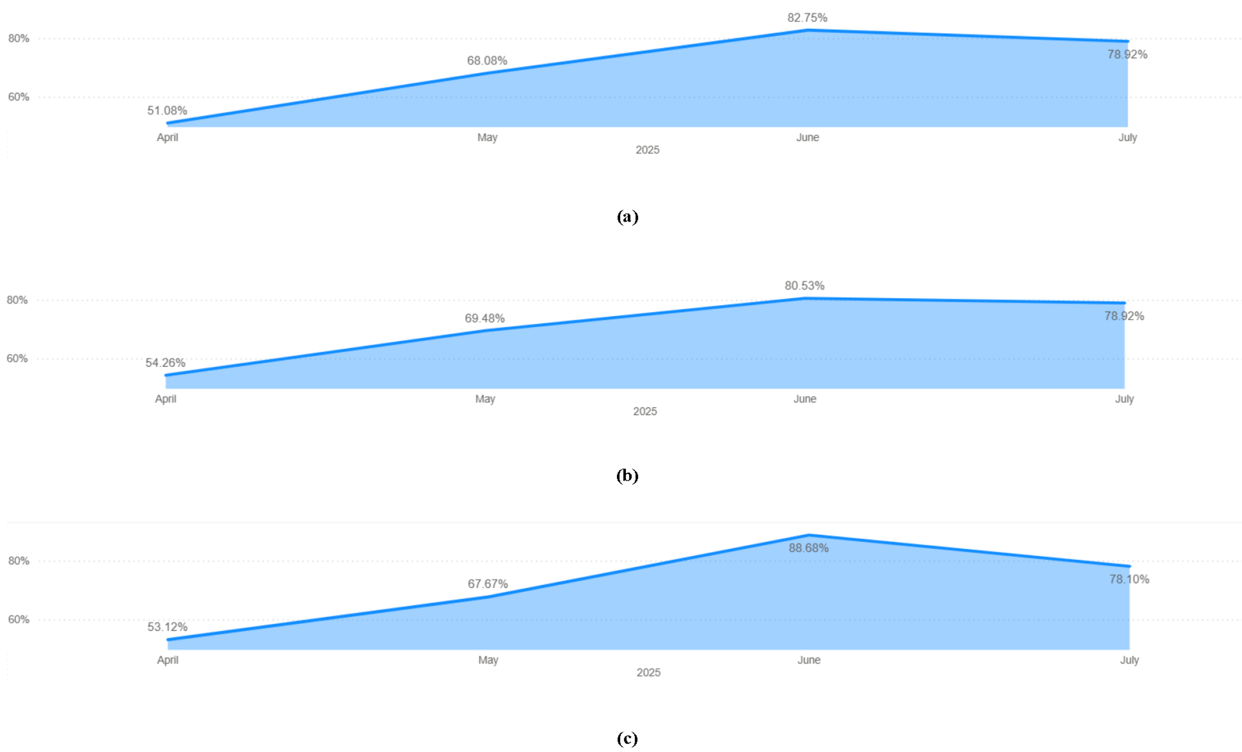


Figure 10—Trend Analysis based on KPIs (a) Total cycle time Trend (b) Sailing Times (c) Loading & Offloading Times

Despite this, analysis of sailing times revealed more consistency, indicating that delays were not attributed to the transit phase but rather to terminal operations. The implementation of the Barge Cycle Monitoring System (BCMS) played a crucial role by enabling real-time data analysis. Previously, such assessments required manual data compilation, causing delays in identifying root causes. With BCMS, stakeholders were able to quickly pinpoint issues, facilitating timely discussions and corrective actions.

Discussion and Conclusion

The implementation of the Barge Cycle Management System (BCMS) yielded significant operational improvements and valuable insights across multiple dimensions of marine logistics. The pilot deployment across a project involving 24 barges over 4 months demonstrated measurable gains in visibility and decision-making.

One of the most impactful outcomes was the introduction of daily tracking and live dashboards, empowering operational teams with timely information on barge location, phase status, loading/offloading progress, and berth occupancy, transforming traditional, fragmented reporting into a unified and responsive control system.

However, the long-term delay analysis revealed that while sailing performance improved, loading and offloading phases continued to face delays, largely due to berth unavailability, identified as a critical bottleneck. Dead time accounted for over 35% of total cycle time, with awaiting berth (18.24%) and weather delays (12.45%) comprising the bulk of non-productive periods. These insights underscore the need for enhanced berth scheduling and better coordination between suppliers and project sites to reduce idle time and improve asset utilization. The average cycle time excluding berth awaiting times, were reduced to 6.6 %.

Over time, only minimal improvements were observed in the overall cycle time, primarily due to persistent congestion at both loading and offloading points. While the sailing phase showed relatively consistent performance. However, this information was available much faster due to the implementation of BCMS, making the response time quicker.

Furthermore, the cost versus performance analysis of tugs uncovered operational inefficiencies, particularly with high-cost vessels delivering suboptimal performance. This evidence-based approach enabled stakeholders to take corrective action, optimize tug selection, and make more strategic hiring decisions moving forward.

Overall, BCMS has proven to be a transformative tool, offering real-time visibility, data-driven analysis, and actionable intelligence. The findings from this implementation provide a solid foundation for broader deployment and ongoing refinement of the sy stem, to drive continuous improvement, cost savings, and operational excellence across all project sites.

Future Work

The Barge Cycle Management System (BCMS) lays a robust foundation for high-integrity data collection and rapid, actionable insights. In the era of AI-driven digital transformation, data quality is paramount, and BCMS ensures that all critical operational data is captured in an analysis-ready format from the outset. This structured approach not only improves operational transparency but also enables swift decision-making at all levels.

Looking ahead, the system has strong potential to evolve into a smart planning assistant. By integrating variables such as material delivery schedules, berth availability, and weather forecasts, BCMS can be extended to generate optimized barge requirement forecasts. These forecasts can account for real-time congestion levels, resource constraints, and delivery priorities, offering dynamic and efficient allocation of marine logistics.

Furthermore, BCMS can be enhanced by integrating with marine traffic systems to automatically detect sailing locations, enabling a hybrid data capture model. This would significantly improve accuracy, minimize manual reporting, and reduce turnaround time for operational insights. As IoT sensors, GPS technologies, and telemetry systems become more widely adopted, BCMS can leverage these tools to further automate reporting processes, ensuring a seamless flow of reliable data across all stages of marine logistics.

To further strengthen operational reliability, the system can incorporate a predictive maintenance module powered by AI. By analysing patterns in barge usage, performance, and environmental factors, the tool could anticipate downtime risks and recommend proactive interventions. Additionally, agentic AI can be implemented as a strategic top layer —capable of summarizing trends, highlighting anomalies, and recommending actions across defined time intervals—thereby accelerating response time and empowering leadership with condensed, yet meaningful intelligence. Ultimately, BCMS is more than just a monitoring tool—it is a scalable digital ecosystem that aligns with the future of autonomous, data-driven marine operations.

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